Predicting Cognitive Stress & Fatigue from Keystroke and Mouse Dynamics

Graham Phillips · George Barakat

DSCI-441 Statistical & Machine Learning · Lehigh University · April 2026 · github.com/georgejbarakat/keystroke-stress-detection

Motivation

- Cognitive stress and fatigue silently degrade decision-making, accuracy, and well-being — especially in knowledge work where they often go unnoticed until significant damage is done.
- Existing measurement tools rely on **intrusive self-report surveys** administered every 30–60 minutes. They are retrospective, disruptive, and miss rapid state changes.
- Every computer already records how a person types and moves their mouse. **Can these passive, zero-hardware signals provide a continuous estimate of cognitive state?**
- Potential applications include workplace wellness monitoring, adaptive interface systems, and longitudinal cognitive health tracking — all with no additional sensors.

Research question: Do features derived from keystroke timing and mouse movement carry sufficient information to classify whether a user is currently experiencing self-reported cognitive stress?

Datasets

Stress Detection by Keystroke & Mouse *(Kaggle)*

- 2 users · 27,233 keystroke events · 5.2M mouse events
- 98 self-reported condition labels (stress, fatigue, energy, pleasantness) collected approximately every 30 min during natural computer use
- Binary stress label: *Stressed* if Stress_Val ∈ {S_Stressed, V_Stressed}; else *Not Stressed*
- Binary fatigue label: *Fatigued* if Fatigue_Val ∈ {Low, Below_Avg}
- **Used for:** supervised training, evaluation, feature engineering

IKDD Keystroke Dynamics *(MDPI / MachineLearnigVisionRG)*

- 374 sessions across 100+ unique users
- Raw inter-key interval (IKI) sequences in milliseconds
- Demographic metadata: gender, age group, handedness, device type
- **Used for:** distributional validation of IKI features and population-level baseline

27K	5.2M	98	374
keystroke events	mouse events	stress labels	IKDD sessions

Label distribution (condition-level):

Statistical Analysis

Mann–Whitney U test (non-parametric, two-sided) on each feature comparing stressed vs. not-stressed windows. Shapiro–Wilk confirmed non-normal distributions, justifying the non-parametric approach.

Feature	U-stat	p-value	Sig?
iki_mean	46,470.5	0.0025	✓
iki_std	46,449.5	0.0025	✓
dwell_mean	37,190.5	0.0950	–
dwell_std	38,269.5	0.2614	–
iki_cv	41,229.5	0.7075	–
typing_speed	40,078.5	0.8359	–
error_rate	39,440.5	0.5935	–

IKI mean and IKI std are the only statistically significant individual features (α = 0.05). Bootstrap 95% CI: Not Stressed IKI mean [450, 510] ms vs. Stressed [520, 600] ms.

Interpretation: Stressed users exhibit longer, more variable inter-key intervals on average — consistent with slower, more effortful typing under cognitive load. However, the effect is small and overlapping, motivating ensemble methods over simple thresholding.

Pipeline & Feature Engineering

Labeling strategy: Each window is assigned the nearest preceding self-report within a 30-minute lookback. Windows without a report within that window are dropped. Final dataset: **464 windows** (268 stressed, 196 not stressed).

Keystroke features (9):

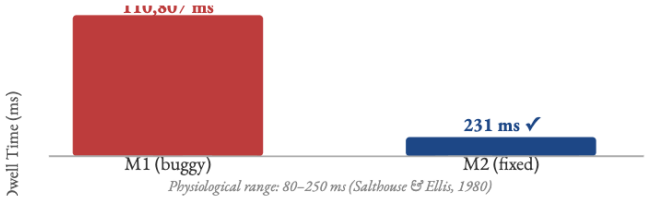
`dwell_mean` `dwell_std` `dwell_median` `iki_mean` `iki_std` `iki_cv` `iki_median` `typing_speed` `error_rate`

Mouse features (4):

`mouse_speed_mean` `mouse_speed_std` `mouse_click_rate` `mouse_move_rate`

Bug Fix: Dwell-Time Outliers (M1 → M2)

M1 bug: Mean dwell time reported as **≈ 110,807 ms** (over 110 seconds). Cause: `Release_Time - Press_Time` was computed across rows that spanned idle gaps between sessions, producing physically impossible values.



Fix: Applied the same outlier filter already used for IKI — drop dwell values above **2,000 ms**. Feature statistics after fix:

Feature	Mean	Median	Std
dwell_mean (ms)	231.5	213.3	97.8
iki_mean (ms)	668.6	675.0	261.2
typing_speed (k/s)	0.46	0.22	1.29
error_rate	0.082	0.069	0.068

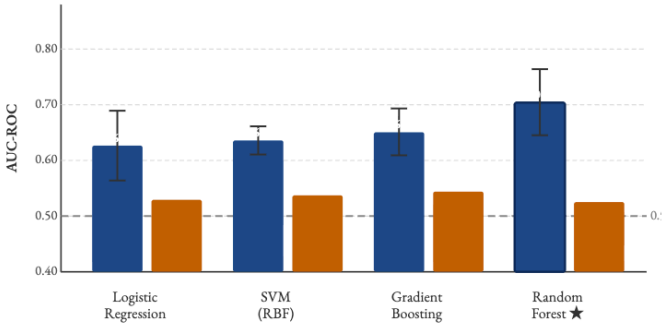
Experiments & Model Selection

Four classifiers, each wrapped in a `SimpleImputer → StandardScaler → Estimator` pipeline, tuned via 3-fold inner grid search, evaluated under two regimes:

- **Random 5-fold stratified CV** — splits data randomly; measures within-dataset discrimination; matches M1 evaluation. Overestimates generalization when users are shared across folds.
- **Leave-one-user-out (LOUO) CV** — each fold trains on one user and tests on the other. Measures true cross-user generalization. Only 2 folds with 2 users.

Model	Hyperparameter grid
Logistic Regression	C ∈ {0.1, 1, 10}
Random Forest	n_estimators=200; max_depth ∈ {None, 8}
Gradient Boosting	n_est ∈ {100,200}; depth ∈ {2,3}; lr=0.1
SVM-RBF	C ∈ {1, 5}; gamma=scale

Evaluation Results



AUC-ROC comparison. Error bars = ±1 std across folds (random CV only). LOUO has 2 folds (2 users).

Model	Random CV AUC	95% CI	LOUO AUC	Δ vs M1
LogReg *(M1 baseline)*	0.627 ± 0.062	[0.572, 0.679]	0.530	—
SVM-RBF	0.636 ± 0.025	[0.612, 0.656]	0.537	+0.009
Gradient Boosting	0.651 ± 0.042	[0.614, 0.686]	0.544	+0.024
Random Forest ★	**0.704 ± 0.059**	**[0.651, 0.754]**	**0.526**	**+0.077**

Best model: Random Forest (200 trees, unconstrained depth, balanced class weights). Beats M1 baseline by **+0.077 AUC** under random CV. Bootstrap 95% CI for RF random CV AUC: **[0.651, 0.754]**.

Key Finding & Honest Limitations

Critical finding: Under leave-one-user-out CV, *all four models* collapse to AUC ≈ 0.53 — near chance. The gap between random CV (0.70) and LOUO (0.53) reveals that the model is primarily learning **per-user typing fingerprints**, not a universal stress signature. The M1 random-split evaluation masked this entirely.

Limitations:

- **n = 2 labeled users** — LOUO has only 2 folds; cross-user conclusions are statistically underpowered
- Self-reports are intermittent and retrospective; ground-truth stress may vary substantially between reports
- Labeling treats stress as piecewise-constant between reports — an acknowledged simplification
- Mouse features are extracted on the same window but may not co-occur with keystroke activity in all windows

Future work:

- **Per-user normalization** — z-score features against each user's rolling baseline to capture deviations from their personal typing style, not absolute values
- **More users** — replicate the labeling protocol with ≥ 20 participants so LOUO evaluation has genuine statistical power
- **Temporal modeling** — LSTM or transformer over sequences of windows to capture state dynamics rather than per-window snapshots
- **Live deployment** — Streamlit app already captures browser-side keystrokes and computes features in real time for immediate inference

0.704	+0.08	0.53
Best AUC (Random Forest)	vs M1 baseline	LOUO AUC (near chance)